XRPL Validator — Daily Engineering Report · 2026-09-14

Period: 2026-09-14 00:00 to 24:00 US Eastern (2026-09-14 04:00 to 2026-09-15 04:00 UTC); ledgers #106,971,914 to #106,994,220. Every count from the ledger and the validator's logs stops at #106,994,220. Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: the day's first and last ledger, from a public full-history server; Step 1 receipt records (they begin 2026-08-29); deployment records (they begin 2026-09-11); xrpl-validator merge records (they begin 2026-07-04); amendment data (carried from the original report). Not available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); signing records (they begin 2026-09-19); Steps 2–5 receipt records (they begin 2026-10-09); data.xrpl.org agreement scores (scored days begin 2026-04-04; some days have none).

Status

Not assessed — no signing records for this day.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** no signing records for this day (they begin 2026-09-19).
- **Native engine:** 26 engine receipts recorded (10 fixed, 0 explained, 16 without a recorded fix).
- **Changes:** 3 deployments (cycles 128, 129, 130), 18 commits merged into main, 11 engine findings fixed (F276–F289).

Service levels

No signing records for this day (they begin 2026-09-19).

Native engine conformance

Ledgers with a receipt in the period (#106,971,914–#106,994,220), by step:

Step	Ledgers with a receipt
Step 1 · native engine	26
Step 2 · records	0

Step	Ledgers with a receipt
Step 3 · order	0
Step 4 · peer data	0
Step 4 · validations	0
Step 5 · own hash	0
Writer	0

Live counters are kept only for the current run; none exist for a past day.

Receipt register

ID	Time	Ledger(s)	Step	Description	Disposition
R-01	—	#106,983,394	Step 1 · native engine	4 ledger change(s) the network made were absent from ours: ECB67430 (deleted), 2A7C2AB2 (deleted), C0B419DF	fixed: F282, commit 28c8adc, deployed in cycle 129 (2026-09-14 21:38 UTC)
R-02	—	#106,983,955	Step 1 · native engine	4 ledger change(s) the network made were absent from ours: C4009D67, 06725EA8 (deleted), 657585B0 (deleted)	fixed: F285–F287, commit 2b4b9ac, deployed in cycle 130 (2026-09-15 01:44 UTC)
R-03	—	#106,984,697	Step 1 · native engine	1 object(s) differed in content	no disposition recorded
R-04	—	#106,986,010	Step 1 · native engine	1 object(s) differed in content	no disposition recorded
R-05	—	#106,986,185	Step 1 · native engine	1 object(s) differed in content	no disposition recorded
R-06	—	#106,988,696	Step 1 · native engine	1 transaction result(s) differed from rippled's; 1 change(s) of ours the network did not make; 1 object(s) differed in content	fixed: F285–F287, commit 2b4b9ac, deployed in cycle 130 (2026-09-15 01:44 UTC)
R-07	—	#106,988,717	Step 1 · native engine	1 transaction result(s) differed from rippled's; 1 change(s) of ours the network did not make; 1 object(s) differed in content	fixed: F285–F287, commit 2b4b9ac, deployed in cycle 130 (2026-09-15 01:44 UTC)
R-08	—	#106,989,075	Step 1 · native engine	1 object(s) differed in content	no disposition recorded
R-09	—	#106,989,105	Step 1 · native engine	1 object(s) differed in content	fixed: F285–F287, commit 2b4b9ac, deployed in cycle 130 (2026-09-15 01:44 UTC)
R-10	—	#106,990,975	Step 1 · native engine	3 object(s) differed in content	fixed: F288, commit 7fe7b60, deployed in cycle 130 (2026-09-15 01:44 UTC)
R-11	—	#106,991,212	Step 1 · native engine	8 change(s) of ours the network did not make	fixed: F289, commit cc5a30e, deployed in cycle 130 (2026-09-15 01:44 UTC)
R-12	—	#106,991,254	Step 1 · native engine	8 change(s) of ours the network did not make	no disposition recorded
R-13	—	#106,992,374	Step 1 · native engine	4 change(s) of ours the network did not make	no disposition recorded
R-14	—	#106,992,486	Step 1 · native	4 change(s) of ours the network did not make	fixed: F290, commit 9718018, deployed in cycle 134 (2026-09-15

ID	Time	Ledger(s)	Step	Description	Disposition
			engine		10:42 UTC)
R-15	—	#106,992,487	Step 1 · native engine	4 change(s) of ours the network did not make	fixed: F290, commit 9718018, deployed in cycle 134 (2026-09-15 10:42 UTC)
R-16	—	#106,992,529	Step 1 · native engine	4 change(s) of ours the network did not make	fixed: F290, commit 9718018, deployed in cycle 134 (2026-09-15 10:42 UTC)
R-17	—	#106,992,532	Step 1 · native engine	4 change(s) of ours the network did not make	no disposition recorded
R-18	—	#106,993,288	Step 1 · native engine	4 change(s) of ours the network did not make	no disposition recorded
R-19	—	#106,993,410	Step 1 · native engine	4 change(s) of ours the network did not make	no disposition recorded
R-20	—	#106,993,481	Step 1 · native engine	4 change(s) of ours the network did not make	no disposition recorded
R-21	—	#106,993,531	Step 1 · native engine	8 change(s) of ours the network did not make	no disposition recorded
R-22	—	#106,993,639	Step 1 · native engine	4 change(s) of ours the network did not make	no disposition recorded
R-23	—	#106,993,640	Step 1 · native engine	4 change(s) of ours the network did not make	no disposition recorded
R-24	—	#106,993,641	Step 1 · native engine	4 change(s) of ours the network did not make	no disposition recorded
R-25	—	#106,993,642	Step 1 · native engine	4 change(s) of ours the network did not make	no disposition recorded
R-26	—	#106,993,793	Step 1 · native engine	4 change(s) of ours the network did not make	no disposition recorded

A receipt is recorded whenever the validator's own work differs from the network's, or the network's trusted validations reached it short of quorum. One ledger flagged by several steps is one receipt. Dispositions: fixed (with the finding, commit and deployment), explained (not a defect, with the reason), or open. Receipt and incident IDs are numbered per report: R-01 here is R-2026-09-14-01, INC-01 is INC-2026-09-14-01. Times are not shown where the reference node no longer holds the ledger.

Incidents

Not assessed: no signing records for this day.

Change log

Deployments

Cycle	Main	Deployed	Test suites	Replay windows	Post-deploy health	Soak
128	`678f5d5`	16:00:47 UTC	n/a	n/a	406 matched, 0 mismatched	#6 from 16:00 UTC: ended early: superseded by the next deployment (2026-09-14 21:38 UTC)
129	`8468b9c`	21:38:35 UTC	n/a	n/a	2675 matched, 0 mismatched	#7 from 21:38 UTC: ended early: superseded by the next deployment (2026-09-15 01:44 UTC)
130	`cc5a30e`	01:44:48 UTC	n/a	n/a	430 matched, 0 mismatched	#8 from 01:44 UTC: ended early: superseded by the next deployment (2026-09-15 04:23 UTC)

Cycle 128 delivered: `03b96c8` fix(xrpl-ledger): F276 — an offer rate whose quotient reaches 2^63 wraps through Number's signed mantissa; `5d804cc` fix(xrpl-ledger): F277 — a "became tiny" or dust reap is the strand's own; only a found-tiny one is permanent; `678f5d5` fix(xrpl-ledger): F278 — the per-iteration fold rests an offer only when its level list accounts for every fill

Cycle 129 delivered: `639e778` feat(ffi,ledger): track 1 — arithmetic oracle exports, differential fuzz harness, two exact primitives; `e7beae9` feat(ledger): port rippled's Number at the 16-digit scale; route IOU add/sub through it (F280b, F281); `94e6904` feat(ledger): AMM drops conversion through Number::to_drops; fuzz STAmount multiply/divide and Number→drops; `a33e1d6` docs: Batch (BatchV1_1) support design — the FFI leg finding, rippled semantics, two-leg plan; `06b21f8` feat(ffi,node): Batch leg A — apply inner transactions inside the outer's shim apply; skip inner ledger entries in the FFI feed; `964a862` feat(node): count inner Batch entries as covered — stats field, metrics line, parity_probe completeness; `6a7f0bd` docs: Batch spec — leg A verification status; `58d8f65` feat(node): vote yes on BatchV1_1; `b26b17b` feat(node): vote yes on fixCleanup3_3_0; `28c8adc` fix(xrpl-ledger): F282 — the bridged crossing's direct strand steps its book by the want, reaping the dead offers it reaches; `8468b9c` fix(xrpl-ledger): F283 — an empty book directory page is no level: the tip, the pool's yardstick and the walk skip it

Cycle 130 delivered: `2b4b9ac` fix(xrpl-ledger): F285/F287 — after a pool-served pass the stream steps the book only while the flow's SendMax budget lasts; F286 — credentials::valid in Payment preclaim; `7fe7b60` fix(xrpl-ledger): F288 — a bridge leg's admission bound reads the raw BookTip, expired or not; `cc5a30e` fix(xrpl-ledger): F289 — a payment's in-driven pass steps the book by its carry, and a fill trimmed on the in side ends the walk

Each deployment passes the unit and port test suites and replays mainnet history windows with zero mismatches before it ships, is health-checked after it starts, and then soaks for 24 hours.

Merged into main

Merged	Repo	Commit	Subject
14:43 UTC	xrpl-validator	`678f5d5`	fix(xrpl-ledger): F278 — the per-iteration fold rests an offer only when its level list accounts for every fill
14:43 UTC	xrpl-validator	`5d804cc`	fix(xrpl-ledger): F277 — a "became tiny" or dust reap is the strand's own; only a found-tiny one is permanent
14:43 UTC	xrpl-validator	`03b96c8`	fix(xrpl-ledger): F276 — an offer rate whose quotient reaches 2^63 wraps through Number's signed mantissa
18:01 UTC	xrpl-validator	`639e778`	feat(ffi,ledger): track 1 — arithmetic oracle exports, differential fuzz harness, two exact primitives
18:01 UTC	xrpl-validator	`e7beae9`	feat(ledger): port rippled's Number at the 16-digit scale; route IOU add/sub through it (F280b, F281)

Merged	Repo	Commit	Subject
18:01 UTC	xrpl-validator	`94e6904`	feat(ledger): AMM drops conversion through Number::to_drops; fuzz STAmount multiply/divide and Number→drops
18:01 UTC	xrpl-validator	`06b21f8`	feat(ffi,node): Batch leg A — apply inner transactions inside the outer's shim apply; skip inner ledger entries in the FFI feed
18:01 UTC	xrpl-validator	`a33e1d6`	docs: Batch (BatchV1_1) support design — the FFI leg finding, rippled semantics, two-leg plan
18:01 UTC	xrpl-validator	`964a862`	feat(node): count inner Batch entries as covered — stats field, metrics line, parity_probe completeness
18:01 UTC	xrpl-validator	`6a7f0bd`	docs: Batch spec — leg A verification status
18:01 UTC	xrpl-validator	`58d8f65`	feat(node): vote yes on BatchV1_1
18:01 UTC	xrpl-validator	`b26b17b`	feat(node): vote yes on fixCleanup3_3_0
18:01 UTC	xrpl-validator	`28c8adc`	fix(xrpl-ledger): F282 — the bridged crossing's direct strand steps its book by the want, reaping the dead offers it reaches
18:01 UTC	xrpl-validator	`8468b9c`	fix(xrpl-ledger): F283 — an empty book directory page is no level: the tip, the pool's yardstick and the walk skip it
00:26 UTC	xrpl-validator	`2b4b9ac`	fix(xrpl-ledger): F285/F287 — after a pool-served pass the stream steps the book only while the flow's SendMax budget lasts; F286 — credentials::valid in Payment preclaim
01:32 UTC	xrpl-validator	`7fe7b60`	fix(xrpl-ledger): F288 — a bridge leg's admission bound reads the raw BookTip, expired or not
01:32 UTC	xrpl-validator	`cc5a30e`	fix(xrpl-ledger): F289 — a payment's in-driven pass steps the book by its carry, and a fill trimmed on the in side ends the walk
02:05 UTC	xrpl-validator	`c089330`	test(xrpl-node): drop a stray brace appended after the F282 vector (gate131 SUITES build error)

Findings fixed

- **fix(xrpl-ledger): F278 — the per-iteration fold rests an offer only when its level list accounts for every fill** (`678f5d5` , xrpl-validator) Finding 274 folded the rested amount over `saved_level_ins` (rippled's per-iteration savedIns). A fill path that spends the taker's whole remaining budget records no level entry — the in side is closed by construction (Finding 30's drain law) and level folding stops — so on an exhausted walk the level list is short of the fills and the fold rests what was never left over. The tail now uses the level fold only when …
- **fix(xrpl-ledger): F277 — a "became tiny" or dust reap is the strand's own; only a found-tiny one is permanent** (`5d804cc` , xrpl-validator) rippled's FlowOfferStream splits the reduced-quality (tiny) removal the way it splits the unfunded one (OfferStream.cpp:308-332): "Removing tiny offer due to reduced quality" is `permRmOffer` when the owner's funds are what the pristine view held, "Removing tiny offer that BECAME tiny due to reduced quality" lives in the strand's sandbox otherwise — discarded with a rejected trial or a killed transaction. Four of …
- **fix(xrpl-ledger): F276 — an offer rate whose quotient reaches 2^63 wraps through Number's signed mantissa** (`03b96c8` , xrpl-validator) `divide` hands `muldiv(numVal, 1e17, denVal) + 5` to `STAmount(asset, uint64, exponent, negative)` , whose `canonicalize()` builds `Number(negative, value_, offset_)` — and Number's mantissa is a signed 64-bit. A quotient at or above 2^63 wraps negative; the magnitude that survives is 2^64 − v, and `getRate` files it with the sign dropped (STAmount.cpp canonicalize / getRate, rippled 3.3.0).

- **feat(ledger): port rippled's Number at the 16-digit scale; route IOU add/sub through it (F280b, F281)** (`e7beae9`, xrpl-validator) Number's mantissa width is amendment-gated (Rules.cpp setCurrentTransactionRules): the 19-digit Large scale switches on with SingleAssetVault or LendingProtocol, neither enabled on mainnet, so transaction processing runs the 16-digit Small scale with the cusp-rounding fix disabled. The arith shim was answering at the process default (Large330) — every "19-digit double rounding" row in the first fuzz was that. …
- **fix(xrpl-ledger): F282 — the bridged crossing's direct strand steps its book by the want, reaping the dead offers it reaches** (`28c8adc`, xrpl-validator) #106983394 34561BC0DC71: rsPrWzpYp5 buys 9862 USDC.rcEGREd8 with RLUSD from a 316.5 RLUSD line, autobridged. rippled's direct strand sizes its BookStep in the rev pass against the want (finding 114): rUgWjzwo's head offer is consumed whole in the sandbox (399.04 USDC), the stream steps to rDeXHakZ's ECB6743009C2 behind it, funded by a zero line — "Removing unfunded offer", a permRmOffer that outlives the pass — and …
- **fix(xrpl-ledger): F283 — an empty book directory page is no level: the tip, the pool's yardstick and the walk skip it** (`8468b9c`, xrpl-validator) #106983394 5B4E06B079D4 (rapido5rxP, RLUSD → USD → CNY → XRP, partial with limit quality): the CNY.rKiCet8/XRP book carries an EMPTY root page at 4D0ADF550834D180 — better than every resting offer and than the pool's spot. Our level list took it as the tip: the pool's turn measured against it ("CLOB better") and never produced the slice, and the walk went on to consume rN5nsAc5's and rwvLbHQ's offers at 4F21… …
- **fix(xrpl-ledger): F285/F287 — after a pool-served pass the stream steps the book only while the flow's SendMax budget lasts; F286 — credentials::valid in Payment preclaim** (`2b4b9ac`, xrpl-validator) Finding 285 (#106983955 D4DD62F77155): rDsxqzXU pays itself 999999999999999 XRP for USD.rKiCet8 from a 29.99 USD line with SendMax 1e15. The pool's slice at the tip's quality takes everything the taker holds; rippled's flow then runs another iteration (remainingIn — the SendMax budget — is still positive), whose rev pass steps the book past the slice and permRmOffers rGSooBxyk's EXPIRED 657585B0C18C on the next …
- **fix(xrpl-ledger): F288 — a bridge leg's admission bound reads the raw BookTip, expired or not** (`7fe7b60`, xrpl-validator) #106990975 36B71A8A3816: rMsXVzCug7's tfPassive 0.02447 BTC → 1907.06654 RLUSD, bridged through the BTC/XRP and XRP/RLUSD pools. rippled admits a strand on `BookStep::tip`, whose CLOB half is `BookTip::step` — the first directory entry as it stands, with no expiry or funding test (the "Removing expired offer" step belongs to the OfferStream and only runs once the strand is flowed). Leg B's raw tip E21DCFAF had …
- **fix(xrpl-ledger): F289 — a payment's in-driven pass steps the book by its carry, and a fill trimmed on the in side ends the walk** (`cc5a30e`, xrpl-validator) #106991212 0B722FF6EA7F: r9tcGwSyYP pays itself 0.071 RLUSD for 50000 drops, XRP → XUSD → USDC.axl → RLUSD, the first two hops pool-served. The last hop's forward pass carries 0.0710127 USDC.axl onto rpkHXWZu's 97.3 head offer. rippled's `fwdImp` consumes an offer whole only while `stpAmt.in <= remainingIn` and stops the moment an offer had to be trimmed to the remaining input (`limitStepIn`, processMore = false): a …

Other engineering changes

- **feat(ffi,ledger): track 1 — arithmetic oracle exports, differential fuzz harness, two exact primitives** (`639e778`, xrpl-validator) The shim gains ffi/xrpl_arith.cpp: one C export per STAmount operation (mulRound, mulRoundStrict, divRound, divRoundStrict, multiply, divide, add, sub, canonicalize), getRate, and Number ops under the four rounding modes — pure functions of their arguments, no ledger. The shim build now matches rippled 3.3.0 (C++23 for std::expected; mpt-crypto from conan for Protocol.h's).

- **feat(ledger): AMM drops conversion through Number::to_drops; fuzz STAmount multiply/divide and Number→drops** (`94e6904`, xrpl-validator) The shim exports Number's explicit int64 conversion (XRPAmount{Number}, `operator rep()`: fraction digits through the guard, the mode decides the drop) and the fuzz puts it, STAmount `multiply` (a Number product at nearest) and STAmount `divide` (the legacy `muldiv(num, 1e17, den) + 5` quotient) through libxrpl and the engine: 31 rows × 20,000 cases, zero mismatches.
- **feat(ffi,node): Batch leg A — apply inner transactions inside the outer's shim apply; skip inner ledger entries in the FFI feed** (`06b21f8`, xrpl-validator)
- **feat(node): count inner Batch entries as covered — stats field, metrics line, parity_probe completeness** (`964a862`, xrpl-validator) Eight of the eleven devnet Batch ledgers 5309212–5309670 probe CLEAN through the outer with every inner folded; the three DIVERGENT ones fail on VaultDeposit / LoanPay hydration (SingleAssetVault / LendingProtocol, devnet-only), not on Batch.
- **feat(node): vote yes on BatchV1_1** (`58d8f65`, xrpl-validator)
- **feat(node): vote yes on fixCleanup3_3_0** (`b26b17b`, xrpl-validator)
- **feat(ledger): track 2 slice 1 — flow module: amounts, QualityFunction, PaymentSandbox, the Step trait** (`1feacf6`, xrpl-validator) The structural port of rippled 3.3.0's path engine begins, behind a module nothing dispatches to yet: - flow::amounts — IOUAmount (a signed 16-digit Number with STAmount's exponent clamp; add/sub are tx::number at ToNearest), XRPAmount, EitherAmount. - flow::quality_function — QualityFunction.h/.cpp: the CLOB-like constant function, the AMM line through the pool balances at the fee, combine, and outFromAvgQ under …

Tests and campaigns

- **test(xrpl-node): drop a stray brace appended after the F282 vector (gate131 SUITES build error)** (`c089330`, xrpl-validator)

Docs and operations

- **docs: Batch (BatchV1_1) support design — the FFI leg finding, rippled semantics, two-leg plan** (`a33e1d6`, xrpl-validator)
- **docs: Batch spec — leg A verification status** (`6a7f0bd`, xrpl-validator)
- **flow: slice 3 — AMMContext, AMMLiquidity::getOffer, AMMOffer, wired into BookStep** (`a2d060a`, xrpl-validator)
- **flow: slice 2 (part 2) — BookStep: forEachOffer, revImp, fwdImp, limitStepIn/Out, consumeOffer** (`01aa737`, xrpl-validator)
- **flow: slice 2 (part 1) — BookTip and FlowOfferStream, TOffer, Quality::ceil_in/out** (`a904cff`, xrpl-validator)
- **flow: slice 1b — STAmount rounding port, pinned by fuzz; slice 2 opens with view.rs** (`c195d83`, xrpl-validator)
- **flow: slice 4 (part 1) — DirectStepI, XRPEndpointStep, rippleCredit/rippleSend/accountSend over the sandbox, XRP deferred credits** (`9743159`, xrpl-validator)
- **flow: slice 4 (part 2) — StrandFlow (single- and multi-strand flow, ActiveStrands, limitOut) and PaySteps (toStrand / toStrands)** (`e0fa870`, xrpl-validator)

- **flow: slice 4 (part 3) — Flow.cpp's flow(), rippleCalculate, and the Payment dispatch behind XRPL_FLOW_ENGINE=port** (`3f5a90e`, xrpl-validator)

Independence program

No independence record for UTC day 2026-09-14; records begin 2026-10-20.

Network

No ledger records for this day: the reference node holds ledgers from 2026-10-04 22:52 UTC (#107,436,647).

Amendment watch

Carried over unchanged from the original report, which read it from the XRP Ledger when it was generated; the reference node no longer holds this day's ledgers.

Read from the XRP Ledger: the Amendments object as of the last ledger of the day, and the day's EnableAmendment pseudo-transactions.

No amendment held a majority.

Amendments enabled on mainnet: 94.

Open items

- Receipt R-03 (#106,984,697): open, under investigation.
- Receipt R-04 (#106,986,010): open, under investigation.
- Receipt R-05 (#106,986,185): open, under investigation.
- Receipt R-08 (#106,989,075): open, under investigation.
- Receipt R-12 (#106,991,254): open, under investigation.
- Receipt R-13 (#106,992,374): open, under investigation.
- Receipt R-17 (#106,992,532): open, under investigation.
- Receipt R-18 (#106,993,288): open, under investigation.
- Receipt R-19 (#106,993,410): open, under investigation.
- Receipt R-20 (#106,993,481): open, under investigation.
- Receipt R-21 (#106,993,531): open, under investigation.
- Receipt R-22 (#106,993,639): open, under investigation.
- Receipt R-23 (#106,993,640): open, under investigation.
- Receipt R-24 (#106,993,641): open, under investigation.
- Receipt R-25 (#106,993,642): open, under investigation.
- Receipt R-26 (#106,993,793): open, under investigation.

Corrections to earlier reports

None.

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **Time-Fair Refilling Iceberg Offers** (dex) Add a tfIceberg flag to OfferCreate plus two new Offer SFields — sfDisplayQuantity (the visible clip, quantized) and sfConcealedQuantity (the hidden r
- **Atomic AMM Flash Loans via Batch** (lending) Add an AMMFlashBorrow inner-transaction type that draws sfAmount of a pool asset directly from an AMM's pseudo-account without burning LP tokens, perm
- **Attestation-Scored Vault Collateral Factors** (lending) Add an AttestationSet transaction and ltATTESTATION ledger object (owner-reserved, keylet::attestation(attestor, subject)) carrying sfScore (uint16) a

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
09-14 14:41	09-14 14:43	xrpl-validator	`678f5d5`	fix(xrpl-ledger): F278 — the per-iteration fold rests an offer only when its level list accounts for every fill
09-13 22:14	09-14 14:43	xrpl-validator	`5d804cc`	fix(xrpl-ledger): F277 — a "became tiny" or dust reap is the strand's own; only a found-tiny one is permanent
09-13 21:48	09-14 14:43	xrpl-validator	`03b96c8`	fix(xrpl-ledger): F276 — an offer rate whose quotient reaches 2^63 wraps through Number's signed mantissa
09-14 15:03	09-14 18:01	xrpl-validator	`639e778`	feat(ffi,ledger): track 1 — arithmetic oracle exports, differential fuzz harness, two exact primitives
09-14 15:40	09-14 18:01	xrpl-validator	`e7beae9`	feat(ledger): port rippled's Number at the 16-digit scale; route IOU add/sub through it (F280b, F281)
09-14 15:56	09-14 18:01	xrpl-validator	`94e6904`	feat(ledger): AMM drops conversion through Number::to_drops; fuzz STAmount multiply/divide and Number→drops
09-14 16:12	09-14 18:01	xrpl-validator	`06b21f8`	feat(ffi,node): Batch leg A — apply inner transactions inside the outer's shim apply; skip inner ledger entries in the FFI feed
09-14 16:12	09-14 18:01	xrpl-validator	`a33e1d6`	docs: Batch (BatchV1_1) support design — the FFI leg finding, rippled semantics, two-leg plan
09-14 16:17	09-14 18:01	xrpl-validator	`964a862`	feat(node): count inner Batch entries as covered — stats field, metrics line, parity_probe completeness
09-14 16:20	09-14 18:01	xrpl-validator	`6a7f0bd`	docs: Batch spec — leg A verification status
09-14 16:30	09-14 18:01	xrpl-validator	`58d8f65`	feat(node): vote yes on BatchV1_1
09-14 16:32	09-14 18:01	xrpl-validator	`b26b17b`	feat(node): vote yes on fixCleanup3_3_0
09-14 16:48	09-14 18:01	xrpl-validator	`28c8adc`	fix(xrpl-ledger): F282 — the bridged crossing's direct strand steps its book by the want, reaping the dead offers it reaches

Committed	Merged into main	Repo	Commit	Subject
09-14 17:00	09-14 18:01	xrpl-validator	`8468b9c`	fix(xrpl-ledger): F283 — an empty book directory page is no level: the tip, the pool's yardstick and the walk skip it
09-15 00:26	09-15 00:26	xrpl-validator	`2b4b9ac`	fix(xrpl-ledger): F285/F287 — after a pool-served pass the stream steps the book only while the flow's SendMax budget lasts; F286 — credentials::valid in Payment preclaim
09-15 01:15	09-15 01:32	xrpl-validator	`7fe7b60`	fix(xrpl-ledger): F288 — a bridge leg's admission bound reads the raw BookTip, expired or not
09-15 01:32	09-15 01:32	xrpl-validator	`cc5a30e`	fix(xrpl-ledger): F289 — a payment's in-driven pass steps the book by its carry, and a fill trimmed on the in side ends the walk
09-15 02:05	09-15 02:05	xrpl-validator	`c089330`	test(xrpl-node): drop a stray brace appended after the F282 vector (gate131 SUITES build error)
09-15 02:05	—	xrpl-validator	`a2d060a`	flow: slice 3 — AMMContext, AMMLiquidity::getOffer, AMMOffer, wired into BookStep
09-15 02:05	—	xrpl-validator	`01aa737`	flow: slice 2 (part 2) — BookStep: forEachOffer, revImp, fwdImp, limitStepIn/Out, consumeOffer
09-15 02:05	—	xrpl-validator	`a904cff`	flow: slice 2 (part 1) — BookTip and FlowOfferStream, TOffer, Quality::ceil_in/out
09-15 02:05	—	xrpl-validator	`c195d83`	flow: slice 1b — STAmount rounding port, pinned by fuzz; slice 2 opens with view.rs
09-15 02:05	—	xrpl-validator	`1feacf6`	feat(ledger): track 2 slice 1 — flow module: amounts, QualityFunction, PaymentSandbox, the Step trait
09-15 02:15	—	xrpl-validator	`9743159`	flow: slice 4 (part 1) — DirectStepI, XRPEndpointStep, rippleCredit/rippleSend/accountSend over the sandbox, XRP deferred credits
09-15 02:19	—	xrpl-validator	`e0fa870`	flow: slice 4 (part 2) — StrandFlow (single- and multi-strand flow, ActiveStrands, limitOut) and PaySteps (toStrand / toStrands)
09-15 02:26	—	xrpl-validator	`3f5a90e`	flow: slice 4 (part 3) — Flow.cpp's flow(), rippleCalculate, and the Payment dispatch behind XRPL_FLOW_ENGINE=port

Times in UTC. A commit appears on the day it was committed to main or merged into it.